CHAPTER 8 PAINT REPAIR PROCEDURE AND INSTRUCTION **8-1**

8.1 GENERAL 8-2
8.1.1 Scope *8-2*
8.1.2 Criterion and code *8-2*
8.1.3 Surface preparation *8-2*
8.1.4 Paint application *8-2*
8.1.5 Paint criterion *8-2*
8.1.6 Inspection *8-3*
8.1.7 Color scheme *8-3*
8.2 PAINT APPLICATION AND REPAIR TECHNICS 8-3
8.2.1 Introduction *8-3*
8.2.2 Surface preparation *8-3*
8.2.3 Shop primer *8-4*
8.2.4 Paint application *8-4*
8.2.5 Inspection *8-6*
8.2.7 repair/touch-up *8-7*
8.3 PAINT INSPECTION AND INSTRUCTION 8-7
8.3.1 Aging or deterioration of coating and judging standard for repair *8-7*
8.4 PAINTING SYSTEM 8-8
8.5 EUROPEAN RUST SCALE 8-10

Chapter 8 Paint repair procedure and instruction

Summary:：

The metal rust is almost in everywhere and it causes very striking harm.According to the statistic: 1/10 of world steel output is wasted because of rust and being eroded every year.

Crane is a steel giant which is always exposed to the salt frog on the seashore. Some of them are alsoexposed to the corrosive cargo (such as the bulk cargo dock). The original antisepsis coat of the crane steel structure will lapse due to the corrosive，which will effect on the equipment appearance .What is the most important is it will lead to eroding the metal structure so as to weaken the load capability even ruin the structure.

The harbor side should bring the routine maintaining into effect while ZPMC makes the paint of the crane steel structure has the appropriate anticorrosive capability and prolongs the time limit of using . The user should update and renew the anticorrosive coat of the crane which is used so as to sustain its function.

The usersintensify the anticorrosive management of the crane, and prolong the life of the coating,which is very important to both safe operation and prolonging the using time limit .It also has an important meaning of saving maintenance costs.

For crane the paint's effect is not only good for look, but also important to preventing steel structure from rusting. Whether the crane structure can reach the maximum designed life is mostly rest with the management and maintenance.

This chapter is mainly about the structure surface treatment, and the criterion of inspection and estimation of painting and restoring the damaged areas during the manufacture procedure.

8.1 GENERAL

8.1.1 Scope

This specification is about both surface preparation, construction technics during the manufacturing and assembling and repair skills of damage area.

This specification is the same with crane or other steel products used ZPMC paint. Without **special instruction, the trademark of paint is ZPMC.**

8.1.2 Criterion and code

This specification refers to the latest edition ASTM, SSPC-SP, and ISO，Table 8-1 list these standards.

Table 8-1：International Standards

Standard	Specification
ASTM A123	Zinc(Hop-Dip Galvanized) Coating on Iron and Steel Products
SSPC-SP-1	Solvent Cleaning
SSPC-SP-2	Hand Tool Cleaning
SSPC-SP-3	Power Tool Cleaning
SSPC-SP-10	Brush-off blast cleaning
SSPC-SP-COM	Surface Preparation Commentary
SSPC-PA2	Measurement of DFT with M-Gange
SSPC-SP-11	Power Tool Cleaning to Bare Metal
ISO 8501	Visual assessment of surface cleanliness
ISO 8502	Test for the assessment of the surface cleanliness
ISO 8502-6	Sampling soluble impurities on surface to be painted
ISO 8503	Surface roughness characteristics of blast-cleaned steel surface substrates
ISO 8504	Preparation of steel substrates before application of paints and related products

8.1.3 Surface preparation

1) Remove all oil and grease with a suitable detergent. Then remove salt and other water-solubility contaminants with high pressure fresh water. Water Pressure should be more than 2,000 psi (138bar). Solvent cleaning shall be done in accordance with SSPC-SP-1.

2) Cleaning all eroded part by power tool should be according to SSPC-_SP-_3.3) Areas adjacent to the damages 50mm shall be feathered back 50mm to assure tight adhesion of the existing paint and new paint.

8.1.4 Paint application

1）Surfaces to be coated shall be clean, dry and dust free before painting Penguard Express MIO.

2） The Penguard Express MIO shall be applied within 4 hours of the time when the surface preparation was over

3）The work condition of individual coats shall be applied in compliance with the conditions that are stipulated in the pertinent JOTON's product datasheets.the work condition should be the same with the requirement in the specification of JOTON paint.

4）specified coat shall be brushed in accordance with JOTON's related code of practice.

5） it is necessary to brushing several times in order to meating need of the specified dry film thickness for each layer.

6）All minimum and maximum recoat intervals in compliance with stipulated in the JOTON's product datasheet must be observed.

7）All corners, welds, sharp edges and areas around bolts and nuts mustbe brushed the primer before the painting.

8.1.5 Paint criterion

1) Exterior：

surface preparation is according to SSPC-SP-3 "Power Tool cleaning" standard，see section 3.

2 × Penguard Express MIO (middle)

1 × Hardtop XP(surface)

2) Interior：

1 × Penguard Express MIO (middle)

3) Galvanized：

Remove all oil and grease according to per8.1.3，

1 × Penguard primer SEA

1 × Hardtop XP

8.1.6 Inspection

1) All work shall beinspected first The inspector shall give notice prior to the start of the surface preparation or coat application so that they canWork under the guide line of inspector.

2) Operation shall not after each inspection points listed below without the approval of the inspector.

-afterCompletion of surface preparationand before primer application;

afterCompletion of application of each coat;

-Final checking and accepting of completedcoat application,

8.1.7 Color scheme

Table 8-2：PAINT COLOR AND SPECIFICATION

COMPONENTS	COLOR
Frame exterior	Red Hempel 54610
Frame interior	RAL 9016
Gantry	Red Hempel 54610
Head block/hook beam	RAL1003
Machinery house exterior	RAL 9016
machinery house/ checker's cabin/ Operator's cab/ Boom hoist checker's cab exterior	RAL 9016
machinery house/ checker's cabin/ Operator's cab/ Boom hoist checker's cab interior	RAL 9016
machinery house/ checker's cabin/ Operator's cab/ Boom hoist checker's cab interior	RAL 7040
operator's cab exterior	RAL 9016
operator's cab interior	RAL 7040

8.2 Paint application and repair technics

8.2.1 Introduction

1）The section covers the surface preparation, materials and coating application requirements of structure equipment and appurtenances of the container crane。

2）The coating system of this specification is based on:

-The international standard ISO 12944 "Protective", under corrosive category "C4"-for coastal or industrial environment;

-With high durability more than 15 years of performance reaching level Ri 3 as defined in ISO 4628-3 or the rust ratio Re 3 defined in European Rust Scale.

3）The color of finish coating is to be in accordance with the owner's requirement, while that of the primer and intermediate coating shall be up to the manufacturer's availability.

8.2.2 Surface preparation

The surface shall be prepared in a manner suitable for coatings. Chemical de-rusters or rust converters are not allowed.

1) Detergent/Solvent Cleaning

Detergent or solvent cleaning shall be performed as outlined in SSPC-SP-1. During the solvent cleaning user should remove visible contaminants with clean duster cloth which is full of solvent .All surfaces to be coated shall be cleaned off all contaminants that would affect the integrity of the coating system. Prior to surface preparation, the surfaces shall be cleaned, as necessary to remove all oil, grease, dirt, salts or other foreign materials. This cleaning shall be done with appropriate detergent or solvent cleaning.

2) Hand or power Tool Cleaning

-Power tool cleaning is a kind of method which to use power tool as polish machine to remove rust and paint

-Once finished, the surface will not be seen any damaged paint. The edge should be eve.

-Please do not polish the surface, the surface should have a certain roughness to ensure the adhesion of the new paint coating.

-When approved, power tool cleaning shall be as per SSPC-SP-3 and SSPC-SP-11. Hand power tool cleaning shall be as per SSPC-SP-2.

-The sharp edges, such as those occurring in rolled structural members as well as those resulting from flame cutting, welding, etc., shall be removed by suitable methods such as a soft grinding wheel before sandblasting and the application of any coat of paint.(Refer to SSPC-SP5 and SSPC-SPCOM).

3) Abrasive Blasting Cleaning

-Prior to abrasive blast cleaning of the steel, remove oil and grease and other contaminations with a suitable detergent / solvent, according to SSPC-SP1"Solvent cleaning".

-All welding seams must have a surface finish that ensures that the quality of the paint system will be maintained in all respects. The sharp edges should be smoothed and the radius of the rounding should be approximately 1mm. holes in welding seam, undercuts,

cracks, residual flux and weld spatter etc. should be avoided. Once it is found they must be remedied by welding and/or grinding.

-Steelwork (including welding, flame-cutting and grinding) must be finished before the surface preparation

-Prior to abrasive blasting, sight glasses, nametags, machinery, electrical fixtures, etc. which are subject to damage from either the blast or the abrasive material remaining shall be protected. If it is possible, nametags should be installed after painting is complete. If it is not possible to tagging after coating, their edges shall be sealed with silicone to prevent moisture accumulation.

-Blasting shall be accomplished so that the previously-painted surfaces will not be contaminated by abrasive or blast wastes.

-Only dry steel grits, silica sand and aluminum silicate grit-type abrasives shall be used (refer to ISO 8504). The abrasive shall comply with government regulations. The abrasive shall be used in accordance with ZPMC's Specification and shall contain no harmful impurities, such as salt, beach sand or rounded river sand is not acceptable.

-All steel surfaces shall be abrasive blasting to follow:

Table 8-3 Steel Surface Blasting Treatment

Item	Description
Cleanliness	"Sa 2.5" according to ISO 8501 or "Near White Blast Cleaning" SSPC SP No.10
Roughness	N9a to N10a (Ra = 6-12.5 micron) Rugotest No.3, Keane-Tator Surface Compactor, 2-3 mils segments (average maximum peak to valley roughness 50-75 micron)
Dust condition	Should be below No.3 scale as per ISO 8504 o

-Surfaces shall not be abrasive blast cleaned when the ambient or the substrate temperature is less than 3°C above the dew point temperature.Blasting cleaned surfaces which show evidence of rust bloom or has been left uncoated overnight, shall be re-cleaned to the specified degree of cleanliness

-Steel surface slivers, laminations, laps, scabs or seamswxposed in any portion of the steel painted by blasting shall be repaired prior to coating. Such area shall be re-blasted prior to coating.

Allowed scour can be used after abrasive blasting cleaning for preventing rust.

- -Welding undercuts and porosities shall be repaired by welding or grinding.Organic filler materials areforbidden.

-When the abrasive blasting is completed, remove all grits and rust.

-Surfaces which areimpracticable to blast clean can be cleaned by power tool or hand tool .Power tool cleaning shall be in accordance with SSPC-SP 11 or 3；lHand tool cleaning shall be as per SSPC-SP-2.

4) Galvanized Surface

- Generally the galvanized surface doesn't need to be painted.

- Some of the crane components which had been galvanized such as rails shall be painted in order tothe better anticorruptive

- Detergent / Solvent cleaning should be performed to remove the surface contaminants. Besides, use the manual / power tool to remove the surface zinc nubble or other edges. Pay attention that do not damage the surface galvanized coating, or the zinc coating should be repaired.

8.2.3 Shop primer

1) Surface preparation

-All steel plates, productsetc. which are to be coated with shop-primer, must be abrasive blasted according to Sa2.5Surface profile should be equivalent to Rugotest No3 N9a to N10a. Also refer to section 8.1.3.

-When the abrasive blasting is completed, remove all grits and dust.

2) When the treated surface is dry and free from oil, dust, etc. the shop-primer Muki Z 2008 should be applied by airless spraying machine immediately on the automatic line.

8.2.4 Paint application

8.2.4.1 .GENERAL APPLICATION REQUIREMENTS

1) Surfaces to be coated shall be clean, dust-free, and dry before the application of any coat of paint and shall meet the requirements for applicable hand, power tool or abrasive blasting surface preparation before priming.

2) The specified coating system primer shall be applied within 4 hours from the moment the surfaces are cleaned to rusting, discoloration, or surface contamination occurs.

3) Outdoor painting shall be done in daylight hours and completed at least one hour before sunset. Indoor painting is allowed 24 hours a day if the specified metal and the air temperature and relative humidity requirements are met inside the building

4) Coating shall be applied under the environmental conditions prescribed in the pertinent JOTON's product

datasheets.

5) No coating shall be applied:

-On wet surfaces,

-When the surface temperature is less than 3°C above the dew point.

-When the ambient or surface temperatures are below the temperature JOTON's recommended for application

-When the wind speed exceeds 7 meters per second.

-Humidity is 80% or more.

6) Coating material shall be finished in the original packing, clearly identifiable, and shall be kept covered,, clean and protected. Materials older than JOTON's recommended shelf lifeare not allowed.

7) All components shall be thoroughly mixed within the time recommended in the JOTON's data sheets. All mixing shall be done in clean containers which are free fromgrease, other paints or contaminants. Containers shall be kept covered to prevent dust, dirt and stain. Paint shall be mechanically mixed in full batches. Partial mixed are not acceptable.

8) Recommended pot life shall not be exceeded. When this limit is reached, the spray pot, hoses, pumps shall be emptied and cleaned. New material should be used.

9) Operation shall be cautious to prevent overspray, spillage or painting the surface to which they are not intended. Skips, sags, and drips are to be repaired.

10) Coatings shall be applied morethan the number of coats specified Individual coat dry film thickness should thicker than the minimum size specified. Wet film thickness needs to be verified at the start of work (until the painter is familiar with the paint and work) with a wet film thickness gauge. Thickness requirements shall be met with each coat and total thickness shall not be calculated by accumulating each coat

11) All coating applied shall be avoided of defects such arsons, sags, pinholes, voids and bubbles.

12) All coatings shall be guaranteed to dry for at least the minimum time recommended, considering the temperature and the humidity, before applying the succeeding coats.

13) Maximum recoating interval shall be as per JOTON's recommendations. When the maximum recoat time has been missed, it needs to perform additional surface preparation. This operation not only includes hand sanding, pressure washing, mist coat application, degreasing or re-blasting.

14) Each coat (primer, intermediate and topcoat) shall be inspected by the yard's Q.C..

8.2.4.2 Spray application

1) The applicators shall be trained with the proper application technique. Proper equipment, as per prescription in the JOTON's datasheets, shall be used.

2) Spray guns, lines, and pressure pots shall be cleaned before adding new material. The air caps, nozzles and needles shall be those recommended by ZPMC for the material being sprayed.

3) Moisture traps or separators shall be installed between the air supply and the pressure pot. Drain valve must be kept open slightly to permit continuous draining of any condensate during operations.

4) All pressure pots shall be equipped with pressure regulators and gauges.

5) Paint shall be applied in a uniform layer, with 30% overlapping of the previous pass. During application, the spray gun shall be held perpendicular to the surface being painted.

8.2.4.3 Roller or Brush Application

1) All welds which cannot be properly spraying coated shall be brush or stripe coated with same coating before applying successive spray operation.

2) Brushes shall be of a style and quality that will enable proper application of the coating.

3) Brushing shall be done so that a smooth coat even in a certain thickness. When applying HB coatings with a brush, two coats may be required to achieve the specified DFT.

8.2.4.4 Application of Inorganic Zinc Primer

1) Inorganic zinc may be applied by either conventional spray application (air spray) or airless spray application. If airless spray is to be used, frequent checks must be taken for the wearing of the parts to ensure proper operation. Inorganic zinc shall be continuously mixed during the application by mechanical agitation.

2) The minimum dry film thickness of inorganic zinc primer shall be 75 microns. In order to prevent mud cracking, the thickness of dry firm can't exceed 130 microns. The defective coating (mud cracking) needs to be removed by blasting and be recoated to the specified thickness. Frequent checks of the film thickness must be made and immediate action must be taken once low DFT is observed.

3) The applicator shall avoid overlapping of inorganic zincwhich isn't necessary. The blast preparation of the uncoated area shall be brought into the existing inorganic-zinc. If the inorganic-zinc has been exposed to the environment and has formed salts or become

contaminated, it should be washed with high-pressure fresh water prior to blast cleaning.

4) Zinc salts as a result of the weathering of inorganic zinc primer should be removed with a fresh water jet before the application of subsequent coatings.

5) When inorganic zinc is over coated with an organic epoxy intermediate, it may be necessary to apply a mist coat of the epoxy paint before applying a full coat, which is helpful for reducing substantially the amount of gassing associated over coating inorganic zinc with epoxies.

8.2.4.5 Application of exterior top coat

1) When the inorganic coating has been inspected and accepted,users carry out general a cleaning by means of dry and clean compressed air blowing method or other methods. The cleaned surface must be dry and free of dust, oil, etc.

2) Carry out stripe coating on all corners, welding seams, sharp edges, hidden areas, etc. by brush application.

3)At all allowed recoating interval, the Penguard Express MIO is applied by airless spray, and the mist coat spraying technique is used on top of the inorganic zinc surface .And then users should build up the required dry film thickness 120 microns as indicated in painting system table 2.

8.2.4.6 Application of exterior top coat

1) When the intermediate coating has been inspected and accepted, carry out a general cleaning by means of dry and clean compressed air blowing method or other methods. The cleaned surface must be dry and free of dust, oil, etc.

2) Carry out stripe coating on all corners, welding seams, sharps edges, hidden areas, etc. by brush application.

3) At the allowed recoating interval, the Hardtop XP top coat is applied by airless spray with a 60 microns dry film thickness as indicated in painting system table 2.

8.2.4.7 Application of interior primer/top coat

1) When the surface preparation has been inspected and accepted, users should carry out a general cleaning by means of dry and clean compressed air blowing method or other methods. The cleaned surface must be dry and free of dust, oil, etc.

2) Carry out stripe coating on all corners, welding seams, sharp edges, hidden areas, etc. by brush application.

3) Apply the Penguard Express MIO by airless spray with 70 microns dry film thickness as indication in painting system table 3.

8.2.5 Inspection

1) All work should be subjected to inspection. The inspectors shall be given notice prior to the start of preparation or coating application so that they can witness the work.

2) Work shall not proceed without the approval of the inspectors at each inspection hold points defined below. The inspector shall ensure that all necessary inspections are carried out at each hold point. The inspection hold points are as follows:

- Completion of surface preparation, beforeprimer application.

- Completion of each coat application (primer, intermediate and top coat).

- Holiday detection and repair.

-Final acceptance of completed coating work.

3) Scaffolding-Beam clamps or any other equipment which will interfere with or damage coating shall not be used. Building yard shall provide safe accesses to all areas for inspections by the inspector.

4) Prior to final coating acceptance, the entire painted surface shall be cleaned with fresh water. If it is necessary, a finish coat may be appliedin the areas stained and contaminated by grinding, weld spatter, general traffic, etc. Areas contaminated by grinding, weld spatter, and other particulate contamination shall be cleaned by hand with sand paper8.2.6 MISCELLANOUS

1) External / Internal Galvanized

-If the zinc coating is damaged, the HEMUCRUL TI-COAT 16490, Grey 19840 should be applied to repair.

- Penguard primer SEA organic epoxy primer coating thickness is 5 microns.

- Applied by airless spray a full coats of Hardtop XP polyurethane finish or similar approved at 50-microns DFT in the shade determined by the owner.

2) Stainless Steel

Stainless steel doesn’t need to be painted

3) Non-Ferrous material

Non- ferrous materials don’t need to be painted.

4) Machined surfaces

Machined surfaces are to be masked prior to painting.

5) Faying surface

The thickness of Resist 78 coating dry film is 50 microns.

8.2.7 repair/touch-up

1) External Steelworks

When the coating of steel surface is damaged, spot cleaning is necessary.

- For areas larger than 0.25 square meters (if the width of the area is over 0.5 meter), the area shall be isolated, blasted to meet the requirement of Sa 2.5 / SSSPC - SP - No. 10. After the metal surface is cleaned, the edge of the surrounding coating shall be feather-edged to remove all the cracked, loose or damaged coatings. Feather-edging shall be accomplished by power tool with a grit wheel or hand-sanding with sandpaper. The prepared surface shall be free of loose, burnt, or blistered coatings. Carry out touch up coating procedure with inorganic zinc followed by intermediate and top coat.

- For areas smaller than 0.25 square meters, use power tool cleaning and apply Penguard Express MIO at 120-micron DFT and followed by top coat.

2) Internal Steelworks

If it is possible that surface preparation must be restored to the degree as originally specified, or as per hand / power tool standards. Apply touch-up coating with airless spray (large areas) or with brush (small areas) Penguard Express MIO at 70-micron DFT. If the application is brush, the paint should be applied in two or more coats to achieve the specified total DFT of 140 microns.

8.3 Paint inspection and instruction

The following passages describe the inspection and repair procedure of the painting system of the crane.

8.3.1 Aging or deterioration of coating and judging standard for repair

1) Rust spreading throughout the steel surface

- In some cases, the coating can be removed with the primer in a wide area. In such case, the rust will eat the steeland make a hole.

-Judgment for repainting: Consider repainting the coating when the defect mentioned above is found.

2) Peeling of the coating

-Peeling of the whole coating due to insufficient primer adhesion to the steel surface.

-Peeling of the top coating due to insufficient adhesion between the top coating and the intermediate coating.

- Peeling of the coating due to cracks.

-Peeling of the coating due to poor painting preparation

Judgment for repainting: If a straight line can be drawn with tip of a drill bit applied on a coating, the coating is considered to be in good condition. If the coating is peeled off, repainting is required.

3) Deterioration of coating

The coating is prone to be removed due to external impacts and internal dissolution of the coating itself.

Judgment for repainting:

-If the coating is vertically and slightly struck with a steel hammer whose diameter is 5-10 mm, it is prone to be removed or be actually removed.

-If the coating is vertically and slightly struck with a steel hammer whose diameter is 5-10 mm, the intermediate coating or the top coating may be removed.

-Such coating is considered to be deteriorated and requires repainting.

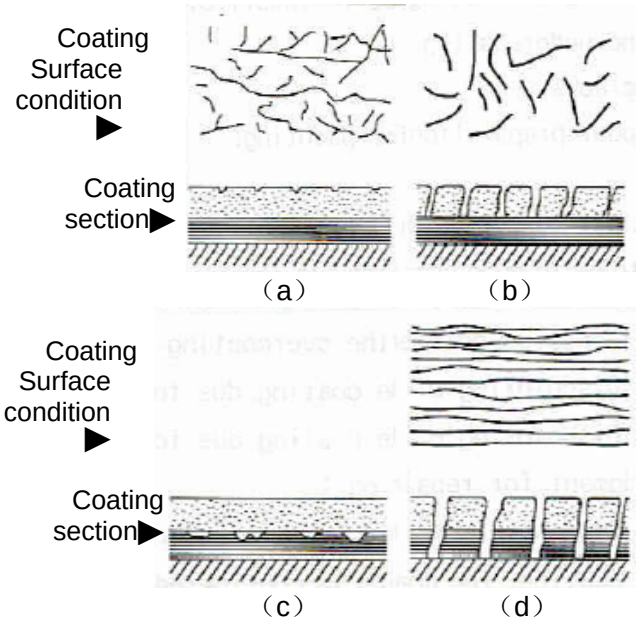
4) Crack in coating

Crack in a coating occurs due to the following causes in the following ways. Refer to Figure A-1 “Crack in Coating” for diagnosis.

-If the crack occurs soon after painting, the cause maybe improper painting or poor painting preparation (Applying top coating before the primer dries up etc.

-The cause of cracks maybe the aging of the coating caused by the shrink and the expansion of the painted steel due to temperature change.

-In some cases, cracks csuddently appeared with sudden temperature change. (Take a special painting way in a cold area where temperature changes frequently).



a: Cracks only on top coating b: Cracks only on top coating c: Cracks only in intermediate coating or primer d: Cracks in both top coating and intermediate/primer.

Figure 8-1：Cracks of the Coating

Judgment for repainting:

- For the State of a and c, No immediate repainting is required according to the view of coating protection. However, the severe cracks may possibly develop into the state D shown in Figure A-1

- For the State of b. The top coating is ineffective and requires re-painting. Re-paint it according to the primer protection effect and the rust condition.

- For the State of d. In most cases, rust also occurs. Repair the painting as soon as possible.

5) Swelling of coating

The causes are considered to be the followings:

In a cold area, the whole coating may be removed in frozen state due to the swelling which once occurred. Refer tophoto 8-2 “Surface View of Swelled Coating”. The swelling could be caused due to:

- Coating soaked in water;

- A drop of water heated in high temperature on the coating;

- Affected by alkali and acid;

- Affected by heat.

Judgment for repainting:

- If swellings occurred in both the top coating and primer lead to rust, repainting is required.

- If the swelling occurs only in the top coating and no rust is found on the steel surfaces, no immediate repainting is required.

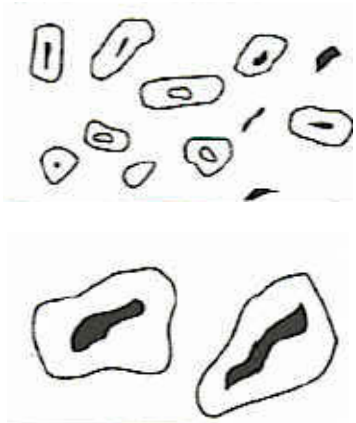


Figure 8-2 Surface View of Swelled Coating

6) Discoloring of coating:

a. The coating may be suddenly discolored due to internal and external causes after painting. In most cases, the causes are improper selection of paint or use of inferior paint.

b. The coating gradually becomes discolored as time passes after paint completion.

Judgment for repainting:

If the coating terribly discolors compared with the original color, it requires immediate repainting.

In case of slight discoloring, no repainting is required.

7) Blurred coating

In general, the coating loses its gloss by 40% in 3-6 months and 60% in 6-12 moths after painting.

Judgment for repainting:

Compare the coating with an original sample. If the coating is only blurred, no repainting is required.

8.4 Painting system

Table 8-1 not only shows the product name and thickness of the primer intermediate and top coating for this crane, but also the sections that need not to be painted.

All values of film thickness are minimum Dry Film Thickness (DFT)!

Table 8-4：Painting system for crane

SHOP PRIMER

Coats	Product name	DFT(microns)
Shop primer	Muki Z 2008	20
Total		20

EXTERNAL AREA

Coats	Product name	DFT(microns)
Primer	Barrier 80	70
Intermediate	Penguard Express MIO	120
Top coat	Hardtop XP	60
Total		270

INTERNAL AREA

Coats	Product name	DFT(microns)
Primer	Barrier 80	70
Top coat	Penguard Express MIO	70
Total		140

GALVANIZED SURFACE (EXTERNAL)

Coats	Product name	DFT(microns)
Primer	Penguard primer SEA	50
Top coat	Hardtop XP	50
Total		100

GALVANIZED SURFACE (INTERNAL)

Coats	Product name	DFT(microns)
Primer	Penguard primer SEA	5
Top coat	Hardtop XP	75
Total		125

REPAIR AREA (A) EXTERNAL AREA (BIG AREA)

Coats	Product name	DFT(microns)
Primer	Barrier 80	70
Intermediate	Penguard Express MIO	120
Top coat	Hardtop XP	60
Total		270

REPAIR AREA (B) INTERNAL AREA (SMALL AREA)

Coats	Product name	DFT(microns)
Intermediate	Barrier 80	70
Top coat	Penguard Express MIO	750

REPAIR AREA (INTERNAL)

Coats	Product name	DFT(microns)
Top coat	Penguard Express MIO	120
Total		120

NO PAINTING PARTS	
1	Rail head
2	Wheel tread / Wheel braking surface if wheel brake used
3	Groove of wire drum and sheave
4	Inside of wire drum
5	Inside of airtight box and pipe
6	Stainless steel and hot-dipped galvanized items (for conduct and conduit fittings)

8.5 European rust scale

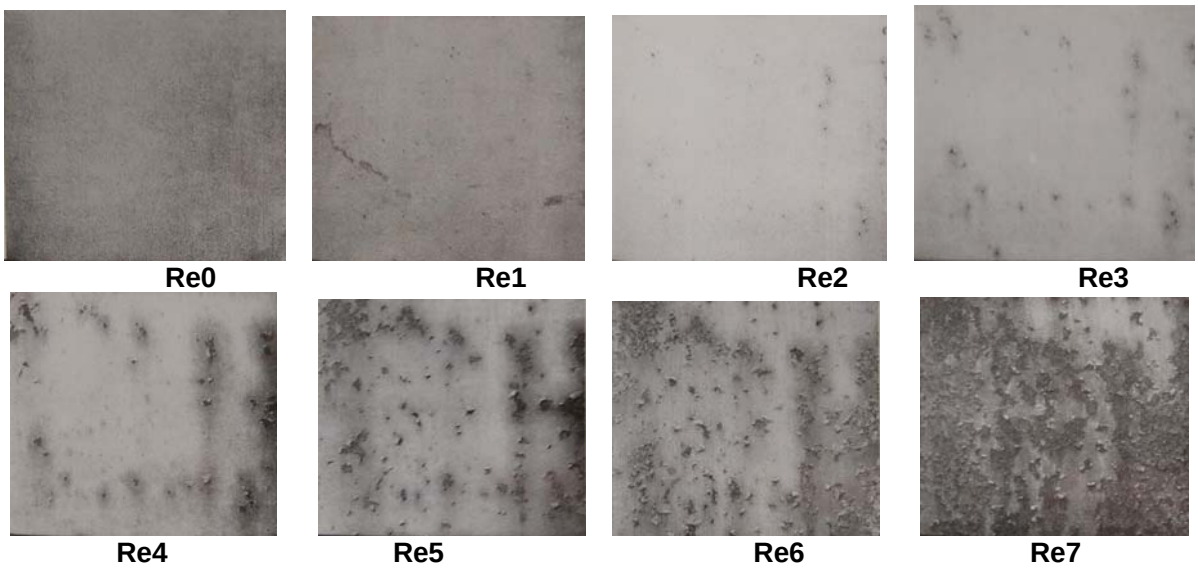


Figure 8-3：Inspection of the swelled coating surface

The European rust scale is divided into 10 stages that shown as Figure 8-3.

The approximate percentages of degree of rusting corresponding to the photo-graphs are shown as follow. The percentages apply only to visible rust which has broken the paint coating. Use the data accompany with the photograph.

Table 8-5：The approximate percentages of degree of rusting

Re1	0.05%	Re5	8%
Re2	0.5%	Re6	15~20%
Re3	1%	Re7	40~50%
Re4	3%		

****